

Oriel summer school

Games with asymmetric information

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July 2024



Outline

1. Games with imperfect information
2. Signaling in the labour market
3. Equilibria
4. Public good game

Games with imperfect information

- Key assumptions in games so far, everyone knows everyone else's payoffs (including their own) for every strategy
- What happens if we relax this assumption?

Games with imperfect information

- Key assumptions in games so far, everyone knows everyone else's payoffs (including their own) for every strategy
- What happens if we relax this assumption?
- These types of games are called Bayesian games. A key aspect of these games is that players are aware that they do not know.
 - They form beliefs about what they do not know from the information they have
 - A key challenge of these games will be to characterise the equilibrium beliefs

Bayesian games

The most widely used games of imperfect information are called Bayesian Games. A Bayesian game is defined by:

- A set of **players**; $N = \{1, 2, \dots, n\}$
- A set of **Action**; $A = \{A_1 \times A_2 \times \dots \times A_n\}$ for all player i
- A set of **types**; $\Theta = \{\Theta_1 \times \Theta_2 \times \dots \times \Theta_n\}$ for all player i
- A **prior**; $p \in f(\Theta)$ which is common to all players
- A **payoff** function; $u_i : \Theta \times A \rightarrow \mathbb{R}$ for all player i

You should already know about the elements in **blue** as they are present in full-info games. The elements in **red** are the novelty here.

Nature the first mover

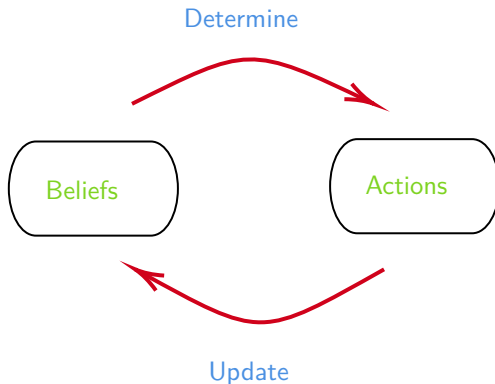
Harsanyi's (1968, Nobel prize-winning) observation:

Incomplete information games can be thought of as games of complete but imperfect information where nature makes the first move, selecting types $\theta_1, \dots, \theta_N$, but not everyone observes nature's move; Player i learns θ_i but not $\theta_{-i} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_N)$.

Types and priors

- In a Bayesian game, players do not know the **type** of other players.
 - But payoffs depend on types so they do not know payoffs either.
 - It is possible to introduce other forms of uncertainty in games but beyond the scope of this class
- You can think of types as any information that is private to player i and relevant to player i 's strategy.
 - Accordingly, a strategy in a Bayesian game is a mapping from type to action; $s_i : \Theta_i \rightarrow A_i$
- In a Bayesian game, players form beliefs about everyone else's type. These beliefs influence their decision-making.

The updating process



An equilibrium in such a game is a situation where all players' beliefs lead them to take the very actions that made them have these beliefs in the first place!

Sequencing terminology

Terminology

1. Ex ante: Before players have learned their types $\theta_1, \dots, \theta_N$
2. Interim: Player i knows own type θ_i but not other players' types θ_{-i}
3. Ex post: Players observe everyone's type $\theta_1, \dots, \theta_N$

Players decide in the interim stage based on the payoff they will get Ex post.

Bayesian nash equilibrium

More formally...

Definition

The game $B(N, A, \Theta, u, p)$ has a Bayesian Nash equilibrium if there is a set of strategy $s^* = \{s_1^*, s_2^*, \dots, s_n^*\}$ such that:

$$E[u_i(s_i^*(\theta_i), s_{-i}^*(\theta_{-i}), \theta_i) | p] \geq E[u_i(s_i(\theta_i), s_{-i}(\theta_{-i}), \theta_i) | p], \\ \forall i \in N, \theta_i \in \Theta_i$$

In plain English, we want to find a set of strategies that give every player the best **expected** payoff given their type.

Same as the best response concept in regular games. But with uncertainty, players cannot know the payoff for sure. So players pick a strategy that gives them the best **expected** payoff instead.

The setting

- Two types of workers: high-ability and low-ability
- Fraction of high-ability workers in the population: λ
- Workers' abilities are unobservable by employers
- Workers can obtain education to signal their ability
- Education **does not** increase ability
- Risk-neutral firms compete to hire workers based on their observed education decisions

Game Progression

1. Workers discover their own abilities
2. Workers decide whether to pursue education ($e = 1$) or not ($e = 0$)
3. Firms observe workers' education decisions **but not** their abilities
4. Firms compete in hiring workers using a Bertrand competition
5. Workers are paid based on their expected productivity

Parameters

- Types of workers: high ability and low ability
- Fraction of high ability workers: λ
- Productivity of high ability workers: y_h
- Productivity of low ability workers: y_l
- Cost of education for high-ability workers: c_h
- Cost of education for low-ability workers: c_l ($c_l > c_h$)
- Education decision: $e = 1$ (obtain education), $e = 0$ (do not obtain education)

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Following the characterisation that we have above, let us identify the set of:

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- **prior**; λ
- **payoff**: wages (workers) and expected profit (firms)

Pooling and separating equilibrium

In Bayesian games we typically have two types of equilibrium:

1. Pooling equilibrium: different types take the same actions
2. Separating equilibrium: different types take different actions

The equilibrium we get depends on the information we have. If there was perfect information, the equilibrium (if any) would be separated. With no information, the equilibrium (if any) would be pooling.

Separating Equilibrium

1. Characterizing a separating equilibrium where education is valued as a signal.
2. Condition for the equilibrium: $y_h - c_h > y_l > y_h - c_l$, where $c_h < c_l$.
3. In the equilibrium, high-ability workers obtain education ($e = 1$), low ability workers choose no education ($e = 0$).
4. Conditional wages: $w(e = 1) = y_h$, $w(e = 0) = y_l$

Best Responses and Signaling

1. Firms: Given workers' strategies, no firm can change behaviour and increase profits.
2. High ability workers: Deviating to no education leads to lower wage $w(e = 0) = y_l$ compared to the current wage $w(e = 1) - c_h = y_h - c_h > y_l$.
3. Low ability workers: Deviating to education leads to higher perceived ability and higher wage $w(e = 1) = y_h$. However, $y_h - c_l < y_l$, making the deviation unprofitable.

Some implications

In this setting, what would happen if there was perfect knowledge?

What would happen if $c_l = c_h = c$?

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What would happen if $c_l = c_h = c$? Again nobody gets educated.
Because the signal is **not** informative. Equivalent to no information. We would have a pooling equilibrium instead and the wage distribution is different (high types get less, low types get more).

Pooling Equilibrium

1. Suppose that firms believe that signals are uninformative
2. Firms are playing best responses. If all workers have the same signal, firms pay the expected productivity of $(1 - \lambda)y_l + \lambda y_h$ (recall that we are assuming perfect competition in the labour market)
3. Wage structure: $w(e = 1) = (1 - \lambda)y_l + \lambda y_h$ and $w(e = 0) = (1 - \lambda)y_l + \lambda y_h$
4. No worker has an incentive to obtain an education

Pooling Equilibrium

- This equilibrium can be sustained if
 - $y_h - c_h < (1 - \lambda)y_l + \lambda y_h$ and $y_l < y_h - c_l$
- Otherwise this equilibrium is "unreasonable"
- This is called the Intuitive Criterion, introduced by Cho and Kreps: Some allocations are technically equilibrium but rely on unrealistic beliefs. Here the belief from firms that all workers are the same is unrealistic if:
 - Strengthened condition: $y_h - c_h > (1 - \lambda)y_l + \lambda y_h$ and $y_l > y_h - c_l$

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- A pooling equilibrium where nobody is educated is more efficient. Why? Because cost of education is a deadweight loss. Total payoffs are higher without education
- But pooling equilibrium makes winners and losers. Who wins, who loses? The high type gets a less than y_h , the low type gets more than y_l . We have cross-subsidisation

The set-up

Public good provision

- Two agents: 1 and 2
- Decide whether to invest in a public good at a cost c_1 , c_2 respectively
- If either 1 or 2 invest, **both** 1 and 2 get a payoff of 1
- Costs of investing c_1 and c_2 are privately known. Costs are either high c^h or low c^l . Probability λ to be c^h

Payoffs	Invest	Do not Invest
Invest	$1 - c_1, 1 - c_2$	$1 - c_1, 1$
Do not Invest	$1, 1 - c_2$	$0, 0$

- Features of Public good:
 - Costly: need at least one agent to invest
 - Envy free (no rivalry): benefits are shared
- Examples of Public goods: national defence, radio, Scientific research, etc.

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- **prior**; λ
- **payoff**: $1, 0, 1-c$

Equilibria

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Suppose that $c^h > 1 > c^l$. Is there a separating equilibrium?

- If the player is high type he never invests
- If the player is low he invests if $1 - c^l > 1 - \lambda$
- We can have a separating equilibrium if $c^l > 1 - \lambda$
- Otherwise no one invests

Implications

- If both players are low and $1 - c^l > 1 - \lambda$ then both players invest $\Rightarrow$ **over-investment**. Why?
- If at least one player is low type and $1 - c^l < 1 - \lambda$ then no-one invests $\Rightarrow$ **under-investment**. Why?
- A great example of how information asymmetry can lead to inefficient outcomes. A twist on the free-rider problem